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4. (a) A class investigated the reaction between magnesium and nitric acid by measuring the volume of hydrogen gas produced over time.

The table shows their results.

Time (s)	Volume of gas produced (cm^3)
0	0
5	12.0
10	20.5
15	29.0
20	33.5
25	35.0
30	35.0

(i) The reaction between magnesium and nitric acid produces magnesium nitrate and hydrogen gas.

Complete and balance the symbol equation for this reaction. [2]

$Mg + \boxed{} HNO_3 \longrightarrow$ .... + ...

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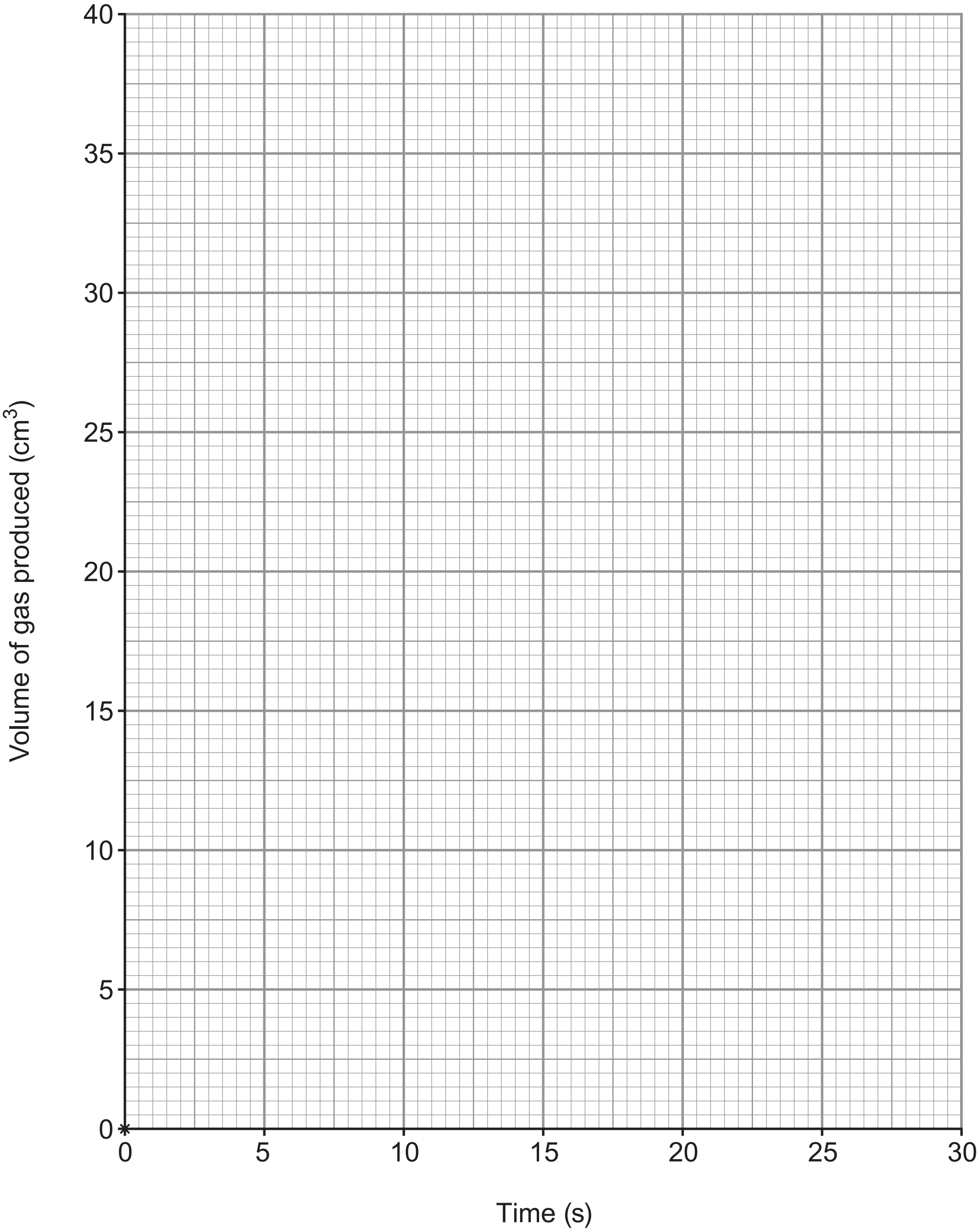
(ii) Plot the results on the grid and draw a suitable line.

The first point has been plotted for you. [3]



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(iii) Use particle theory to explain why the graph becomes less steep over time. [3]

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(b) State why a catalyst is added to a reaction mixture and explain how a catalyst works. [2]

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